

The Rules of Mathematical Typography

A LaTeX field guide for documents that don't look like LaTeX defaults

Good typography is invisible. Nobody finishes a paper thinking “what lovely thin space kerning before that differential.” They just don't notice anything *wrong*. Every rule below exists to remove a small, nagging visual error you have trained yourself not to see anymore.

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1 Constants, Operators & Differentials

1. Upright constants and operators

Mathematical constants and named operators are not variables, don't slant them like one.

`$e^x$, $i$, $\int f(x)dx$`
 e^x , i , $\int f(x) dx$

`\mathrm{e}^x, \mathrm{i}, \mathrm{d}x`
 e^x , i , $\int f(x) dx$

2. Correct spacing around differentials

A thin space ($\,$) separates the integrand from dx so the eye reads them as two distinct objects.

Cramped: $\int f(x)dx$ `\int f(x)\mathrm{d}x`

Open: $\int f(x) dx$ `\int f(x)\,, \mathrm{d}x`

3. Use proper operator names

$\sin$, $\log$, $\lim$ typeset as three italic variables multiplied together unless declared as operators.

`$\sin(x)$` → $\sin(x)$ (s, i, n – three variables)

`$\sin(x)$` → $\sin(x)$ also: `\log`, `\ln`, `\exp`, `\lim`, `\max`, `\min`

4. Distinguish transpose from exponent

If T means *transpose*, it isn't a variable, so don't italicize it.

$A^T \rightarrow A^T$ (ambiguous) vs. $A^{\top} \rightarrow A^{\top}$ (unambiguous)

5. Use the correct glyph for primes

Never type an apostrophe in text mode and expect a derivative mark.

`f'` (text mode) produces a typewriter apostrophe, not a prime

`$f'$, $f''$, $f^{(n)}$` f' , f'' , $f^{(n)}$ – true prime glyphs

6. Don't use italic capitals for common sets

$\mathbb{R}$, $\mathbb{C}$, $\mathbb{N}$, $\mathbb{Z}$ read instantly as *sets*, distinct from any italic variable R , C , N , or Z in the same document.

2 Spacing & Delimiters

7. Scale delimiters only when needed

`\left( \right)` on every parenthesis makes simple expressions visually noisy.

$(a + b)$	$(a+b)$	needs nothing
$\left(\frac{a+b}{c+d}\right)$	<code>\left(\dfrac{a+b}{c+d}\right)</code>	needs scaling

8. Use proper minus signs

A minus sign only exists in math mode. A text hyphen is not a substitute.

Always write subtraction inside `$...$`: `$x-y$` $\rightarrow x - y$. LaTeX renders the correct glyph automatically once you are in math mode.

9. Avoid inline fractions when possible

`\frac` inside a paragraph distorts line spacing. A slashed form is often kinder to the page.

Inline $\frac{a+b}{c+d}$ stretches the line vs. inline $(a+b)/(c+d)$ stays flush.

10. Classify custom delimiters

TeX spaces symbols differently depending on whether they're ordinary, binary operators, relations, openings, closings, or punctuation. Custom notation should declare its class with `\mathopen{}` / `\mathclose{}` so spacing stays correct.

11. Use `\colon` instead of `:`

<code>f:A\to B \rightarrow f:A \rightarrow B</code>	cramped
<code>f\colon A\to B \rightarrow f:A \rightarrow B</code>	proper mathematical spacing

12. Don't use `\cdot` for everything

Different products carry different meanings.

$a \cdot b$ ab $a \times b$ $a \otimes b$

13. Use ellipses correctly

`...` is a typewriter artifact, not a mathematical ellipsis.

`\ldots`, `\cdots`, `\vdots`, `\ddots`
 $a_1, a_2, \dots, a_n$ $A_1 A_2 \cdots A_n$

14. Distinguish text and math punctuation

A comma between two math expressions needs the same spacing rules as in prose.

$f(x), g(x)$ $f(x), g(x)$ correct
 $f(x), g(x)$ with a manual space usually redundant / inconsistent

15. Thin spaces in numbers

1000000 vs $1\,000\,000 \rightarrow 1\,000\,000$ or $\text{\num{1000000}}$ (via siunitx) $\rightarrow 1\,000\,000$

16. Align equations on meaning, not appearance

Readers scan equals signs vertically, give them a column to scan.

$a+b=c$
 $x+y=z$

```
\begin{align}
a+b &= c \\
x+y &= z \\
\end{align}
```

3 Units & Notational Consistency

17. Distinguish vectors from scalars consistently

Pick one convention per document and never mix it.

$\mathbf{v}$ or $\vec{v}$ or v the choice matters less than the consistency

18. Use roman text for units

$10\textit{km}$ (10 km, italic, cramped) bad
 10 km ($10\,\mathrm{km}$) better
 $\text{\SI{10}{kilo\meter}}$ $\rightarrow 10\text{ km}$ best, via siunitx

19. Use semantic commands

Don't repeat $\mathbb{R}$ fifty times.

```
\newcommand{\R}{\mathbb{R}}
```

Then write $\R$ everywhere: $\mathbb{R}, \mathbb{C}, \mathbb{N}, \mathbb{Z}$. Trivial to redefine globally later, and the document above does exactly this.

20. Match notation to discipline

Typography is partly about fitting the conventions of your readers.

Physics: $\partial_t u$ Mathematics: $\frac{\partial u}{\partial t}$ Engineering: $\dot{x}$

4 Punctuation, Dashes & Quotes

21. Use proper quotation marks

"quote"

``quote'' → “quote”

straight quotes, wrong in prose
correct typographic quotes

22. Pay attention to dashes

Three different characters, three different jobs.

-	hyphen	co-author
--	en dash	pages 10–20
---	em dash	He paused—then continued.

23. Use non-breaking spaces

A tilde (~) glues two tokens so TeX never splits them across a line break.

Fig. ~3, Eq. ~(5), Dr. ~Smith, Theorem~2

24. Never manually insert line breaks to fix layout

`\` scattered through prose fights TeX's own justification engine and tends to create worse spacing later in the document, not better.

25. Proper quotes inside code listings too

If code is shown as prose rather than executed, curly quotes can look better than straight ones. Reserve straight quotes for code that is meant to actually compile or run.

5 Document-wide Polish

26. Don't abuse boldface

Hierarchy comes from whitespace and structure, not from bolding every other sentence.

27. Use microtype

One of the largest visual upgrades available for almost no effort.

```
\usepackage{microtype}
```

Enables character protrusion, font expansion, and better justification. Most readers won't know why a document looks better. It just does.

28. Care about widows and orphans

A *widow* is a paragraph's last line stranded alone at the top of the next page; an *orphan* is a paragraph's first line stranded alone at the bottom of the previous page. Avoid both.

29. Use proper theorem styling

A theorem should *look* different from a definition.

Theorem	italic body
Definition	upright body
Remark	upright body

Once upright differentials, thin spaces before dx , real operator names, `siunitx` units, and disciplined delimiter usage become habits, typographic mistakes in ordinary textbooks start jumping off the page. It's hard to unsee.